Hospital Readmission Risk Analysis

SQL case study | 69,973 diabetic patients | 130 U.S. hospitals | 1999-2008

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69,973	9.0%	3.6x	3.8x
patients in the final analytic cohort	overall 30-day readmission rate	higher risk with 4+ prior inpatient visits (29.3% vs 8.1%)	higher risk when discharged to rehab vs home (26.3% vs 6.9%)

Executive Summary

Hospitals are financially penalized when patients return within 30 days of discharge. Using **MySQL**, I cleaned and analyzed 101,766 diabetic hospital encounters to find **which patients are most likely to be readmitted** and where care teams should concentrate follow-up effort.

> **Bottom line:** Two simple, data-available signals, **prior inpatient history** and **discharge destination**, flag just **20.7% of patients** who account for **36.7% of all 30-day readmissions** (a 15.9% readmission rate, 1.8x the baseline). That is a practical, low-cost targeting rule for discharge teams.

Key Findings at a Glance

- **Prior utilization is the strongest signal.** Readmission climbs steadily from 8.1% (no prior inpatient stays) to 29.3% (4+ stays).
- **Discharge destination matters.** Patients sent to rehab (26.3%), another inpatient institution (20.6%) or a short-term hospital (13.8%) return far more often than those sent home (6.9%).
- **Diagnosis category:** injury (10.8%) and circulatory (9.7%) diagnoses carry the highest rates; diabetes as the primary diagnosis sits just above average (9.1%).
- **Complexity compounds risk:** longer stays, heavier medication burden (21+ drugs: 10.5% vs 7.3%) and older age all push risk upward.
- **Medication changes and A1C testing** show smaller, statistically detectable differences, useful context but weaker levers than utilization history.

Business Problem

Under value-based care programs, unplanned 30-day readmissions cost hospitals money and signal gaps in discharge planning. The question this project answers: **who is at highest risk, what patterns drive it, and what should a care team do differently?**

Tools & Techniques

Area	What I used
Database	MySQL Workbench
SQL techniques	CTEs, window functions (RANK, NTILE), CASE-based risk tiering, correlated subqueries with HAVING, multi-table joins, self-join deduplication
Dataset	UCI "Diabetes 130-US Hospitals for Years 1999-2008" (101,766 encounters, 50 attributes)

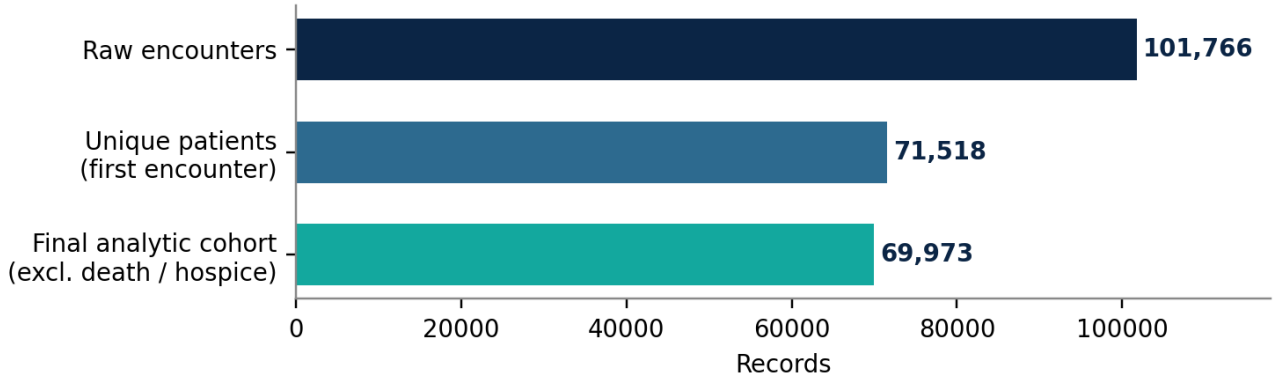
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1. Data & Cleaning Methodology

Raw hospital data is messy, and the cleaning decisions change the answer. Every step below was done in SQL and was chosen to avoid a specific bias.

Step	SQL action	Why it matters
1. Standardize missing values	`UPDATE ... SET col = NULLIF(col, '?')` on race, weight, payer_code, specialty, diag_1-3	The dataset encodes "unknown" as "?" which silently breaks counts and averages
2. Profile missingness	Computed % NULL per column	Weight 96.9% missing, medical specialty 49.1%, payer code 39.6%, race 2.2%. Weight was excluded from analysis as unusable
3. One row per patient	Self-join on `MIN(encounter_id)` per patient_nbr into `diabetic_data_dedup`	101,766 encounters belong to only 71,518 patients. Repeat visits by the same person are not independent and would inflate rates
4. Remove patients who cannot be readmitted	Deleted discharge codes 11, 13, 14, 19, 20, 21, 26 (expired, hospice, unknown)	A patient who died or entered hospice cannot return, so keeping them understates risk. Cohort fell by 1,545 patients
5. Decode IDs	Built `discharge_disposition_map` and `admission_source_map` lookup tables	Turns numeric codes into readable labels for business reporting
6. Feature engineering	Added `age_midpoint` via CASE on age bands	Enables ordered age analysis and window-function ranking

From raw data to analytic cohort



Why this matters: without deduplication and the death/hospice exclusion, readmission rates would be biased toward frequent visitors and understated for survivors. Documenting these decisions is what separates a trustworthy analysis from a quick query.

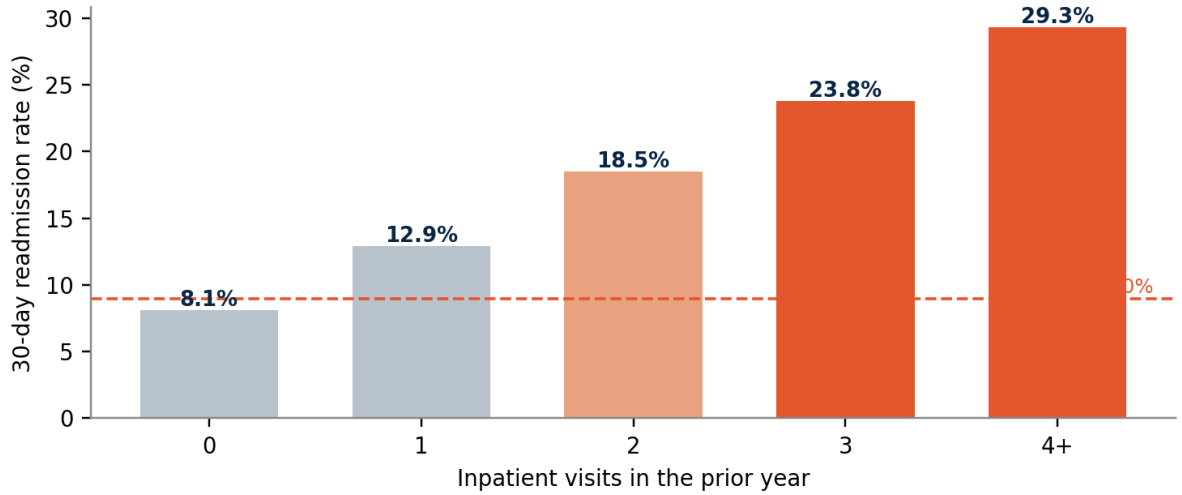
Outcome definition

Readmitted within 30 days (readmitted = "<30") is the target event. The cohort split is 59.3% not readmitted, 31.8% readmitted after 30 days, and **9.0% readmitted within 30 days**, which is the baseline used throughout this report (dashed line in every chart).

2. Who Is Most Likely to Come Back?

Finding 1: Prior inpatient history is the strongest predictor

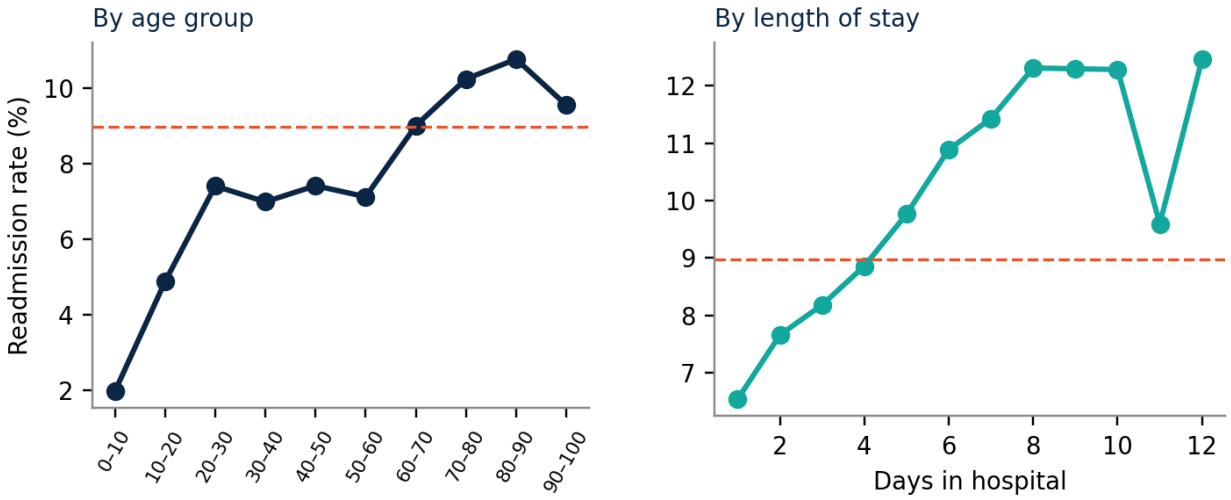
Each additional inpatient stay in the prior year raises the readmission rate in a clear, stepwise pattern. Patients with **four or more** prior stays are readmitted at **3.6x** the rate of patients with none.



Prior inpatient visits	Patients	30-day readmission rate
0	61,782	8.1%
1	5,794	12.9%
2	1,501	18.5%
3	463	23.8%
4+	433	29.3%

Business insight: patients with 2+ prior inpatient stays are only **3.4%** of the cohort (2,397 people) but generate **8.2%** of all readmissions. A chi-square test confirms the relationship is highly significant ($p < 0.001$).

Finding 2: Age and length of stay add context

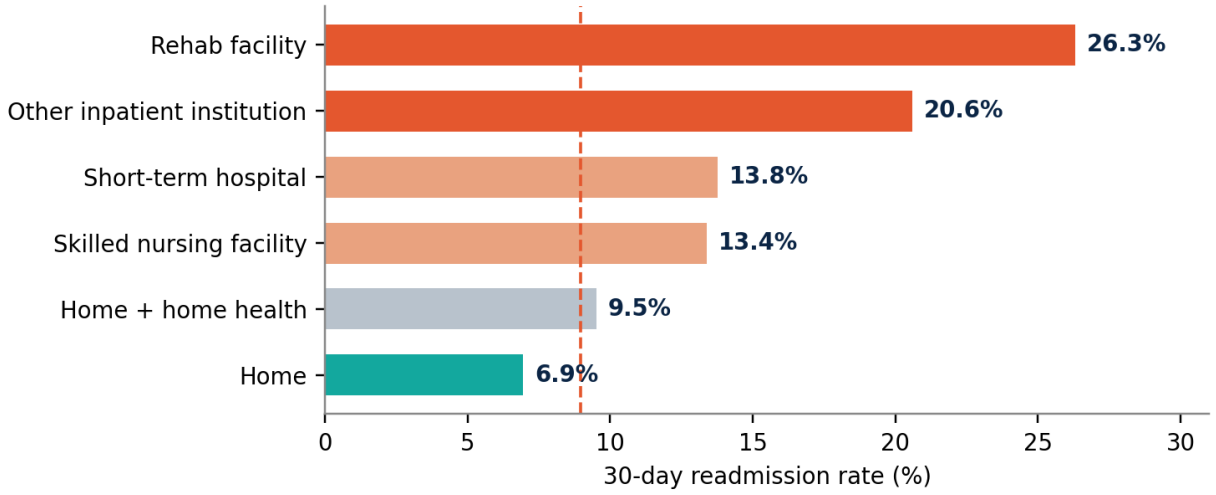


Risk rises from 2.0% in children to a peak of **10.8% at ages 80-90**, and from 6.5% after a one-day stay to roughly **12% after 8+ days**. Both reflect higher clinical complexity, and both are recorded at admission, so they are available early for risk flagging.

3. Where Patients Go After Discharge Matters

Finding 3: Discharge disposition

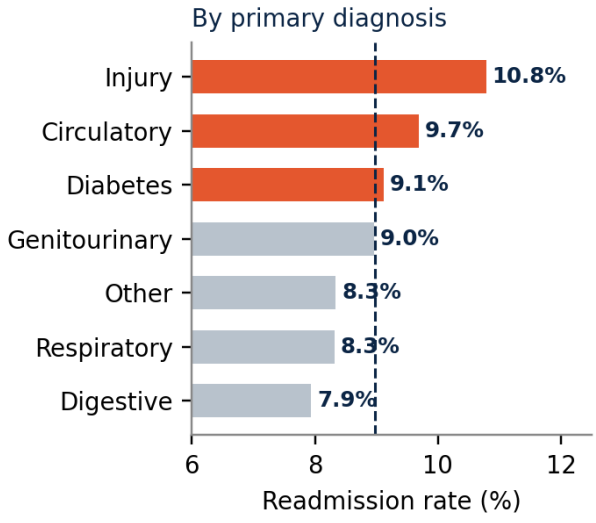
Joining encounters to the `discharge_disposition_map` lookup table reveals a sharp split between patients discharged home and those transferred to another care setting.



Business insight: patients discharged to rehab, skilled nursing, another inpatient institution or a short-term hospital make up **18.1%** of the cohort but account for **31.0%** of 30-day readmissions (15.4% rate vs 6.9% for home). These handoffs are where discharge communication and follow-up can fail.

Small groups such as psychiatric-hospital transfers (35.6%, n = 90) show very high rates but are excluded from the chart because the sample is too small to be reliable. The SQL query applies `HAVING COUNT(*) > 100` for this reason.

Finding 4: Primary diagnosis category



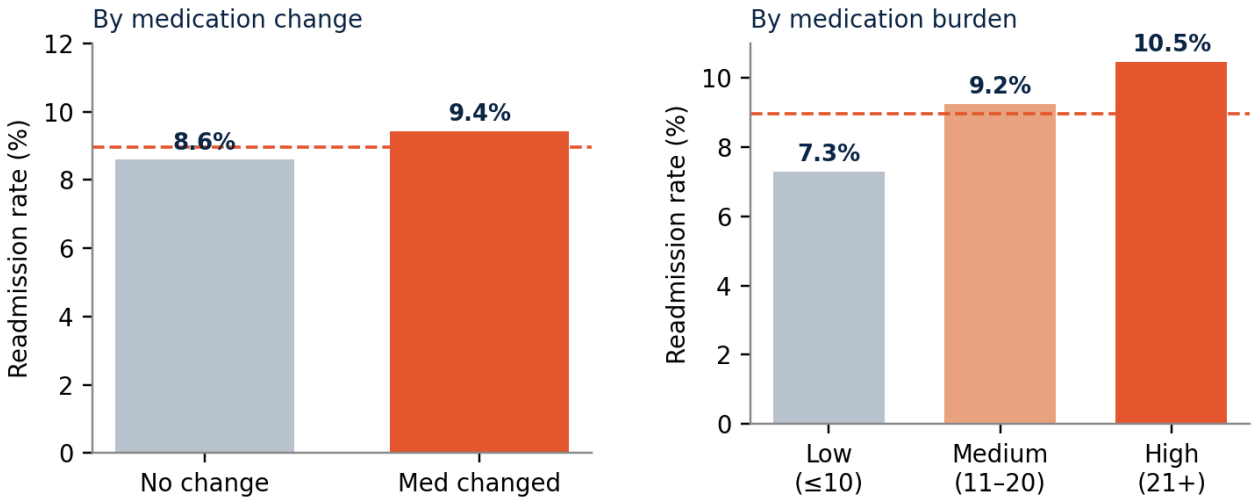
Using a CASE expression on ICD-9 code ranges, then a **correlated subquery with HAVING**, I isolated the diagnosis categories that perform **above the population average** of 9.0%.

- **Injury & poisoning:** 10.8%, highest
- **Circulatory:** 9.7%, and the largest group above average (21,316 patients)
- **Diabetes (250.xx):** 9.1%, marginally above average
- **Genitourinary:** 9.0%, in line with average
- **Digestive 7.9%** and **respiratory 8.3%** fall below average

Differences across categories are statistically significant ($p < 0.001$) but modest in size, so diagnosis is best used as a secondary signal after utilization history and discharge destination.

4. Clinical Process Signals & SQL Craft

Finding 5: Medication changes, medication burden and A1C testing



- **Medication change during the stay** is associated with a higher rate (9.4% vs 8.6%, p < 0.001). This likely reflects sicker patients needing adjustments rather than harm from the change itself.
- **Medication burden** shows a dose-response pattern: 7.3% (up to 10 drugs), 9.2% (11-20) and 10.5% (21+).
- **A1C testing:** patients who had an A1C test during the stay were readmitted slightly *less* often (8.4%) than those untested (9.1%; p = 0.012), consistent with the original clinical research question about glycemic management.

> **Analytical honesty:** these are unadjusted, observational rates. They show *association*, not causation, and the effects are smaller than those of prior utilization and discharge destination.

SQL Techniques Demonstrated

Technique	Used for
CTE (`patient_risk_base`)	Staged a reusable cohort with a 30-day readmission flag across several queries
`RANK() OVER (PARTITION BY age_midpoint)`	Ranked medication burden within each age group
`NTILE(4)`	Split patients into risk quartiles by prior inpatient count
CASE tiering	Converted medications and diagnosis counts into Low / Medium / High business tiers
Correlated subquery + HAVING	Kept only diagnosis categories above the population-wide average
Self-join + `MIN()`	Deduplicated to one encounter per patient
Multi-table joins	Translated numeric IDs into readable admission and discharge labels

Sample: above-average diagnosis categories

```
WITH diag_categorized AS (
  SELECT encounter_id,
    CASE WHEN diag_1 LIKE '250%' THEN 'Diabetes'
         WHEN CAST(LEFT(diag_1,3) AS UNSIGNED) BETWEEN 390 AND 459 THEN 'Circulatory'
         ... ELSE 'Other' END AS diagnosis_category,
    CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END AS is_readmitted_30
  FROM diabetic_data_dedup WHERE diag_1 IS NOT NULL)
SELECT diagnosis_category, COUNT(*) AS total_encounters,
       ROUND(AVG(is_readmitted_30)*100,1) AS readmission_rate_pct
FROM diag_categorized GROUP BY diagnosis_category
HAVING AVG(is_readmitted_30) > (SELECT AVG(is_readmitted_30) FROM diag_categorized)
ORDER BY readmission_rate_pct DESC;
```

5. Recommendations, Limitations & Next Steps

Recommendations for care teams

#	Action	Evidence
1	**Flag high-risk patients at discharge:** any patient with 2+ prior inpatient stays or a transfer to rehab, SNF, or another inpatient setting gets a priority follow-up call within 48-72 hours	20.7% of patients, 36.7% of readmissions, 15.9% rate (1.8x baseline)
2	**Strengthen handoffs to facilities:** standardize discharge summaries and medication reconciliation for transfers; audit facilities with the highest return rates	Rehab 26.3%, other inpatient 20.6%, SNF 13.4% vs home 6.9%
3	**Extend care coordination for circulatory and injury cases** and for patients on 21+ medications (pharmacist review)	Injury 10.8%, circulatory 9.7%, 21+ meds 10.5%
4	**Make A1C testing and medication changes part of the discharge checklist,** with documented follow-up when meds change	A1C tested 8.4% vs 9.1%; med change 9.4% vs 8.6%

Limitations (what I would tell a stakeholder)

- **Observational data:** results show association, not causation, and rates are unadjusted for other factors.
- **Historical data (1999-2008):** care patterns have evolved, so thresholds should be revalidated on current data.
- **First encounter only:** keeping one row per patient avoids bias but excludes later visits, so it cannot show how risk changes over time.
- **Diabetic population only,** so findings may not generalize to all hospital patients.

Next steps

- Build a **logistic regression / gradient-boosting risk score** in Python and compare it with the rule-based tiers above.
- Publish the findings as an interactive **Power BI dashboard** with drill-downs by diagnosis, age and discharge setting.
- Validate the targeting rule with a cost-benefit estimate of avoided penalties per follow-up call.

Skills Demonstrated

Technical	Analytical	Communication
Advanced MySQL	Cohort definition	Business framing
Data cleaning & deduplication	Bias awareness	Insight-first storytelling
Window functions & CTEs	Statistical testing (chi-square)	Actionable recommendations
Relational lookup design	Risk segmentation	Transparent limitations

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Full SQL code and documentation: https://github.com/jeeruvicky51-bit/Hospital-Readmission-Risk-Analysis